



AI DATA ANALYST AGENT

Professional Data Analysis Report

Generated On: 25 July 2026 06:23 PM

Developer: Pritiranjana Rout

Version: 2.0.0

EXECUTIVE SUMMARY

This report provides a comprehensive evaluation of the customer transaction dataset, which consists of 99,308 records. The dataset exhibits a high overall health score of 90%, with zero missing values and zero duplicates. Despite these strong quality metrics, the presence of 31,499 potential outliers and a significant portion of churned customers presents both risks and opportunities. Strategic intervention is required to address anomalies and optimize customer retention.

REPORT INFORMATION

Property	Value
Generated By	AI Data Analyst Agent
Developer	Pritiranjana Rout
Version	2.0.0
Generated On	25 July 2026 06:23 PM

DATASET SUMMARY

Metric	Value
Rows	99308
Columns	10
Health Score	90%
Duplicate Rows	0

DATASET HEALTH SCORE

Overall Health Score : 90%

MISSING VALUES

Column	Missing
customer_id	0
age	0
annual_income	0
credit_score	0
spending_score	0
account_balance	0
department	0
education	0
is_churned	0
transaction_date	0

OUTLIER ANALYSIS

Column	Outliers
age	1935
annual_income	41
credit_score	0
spending_score	0
account_balance	4755
is_churned	24768

STRONG CORRELATION ANALYSIS

Column 1	Column 2	Correlation
No Strong Correlation	-	-

DATASET OVERVIEW

The dataset contains 99,308 rows and 10 columns capturing customer demographic and financial profiles. The tracked attributes include customer ID, age, annual income, credit score, spending score, account balance, department, education, churn status, and transaction date. These variables span multiple organizational units, including Finance, HR, and Sales, and cover diverse educational backgrounds such as High School, Bachelor, and PhD.

DATA QUALITY ANALYSIS

The dataset demonstrates excellent completeness with a health score of 90%. There are no missing values or duplicate records across any of the columns. However, data quality is impacted by 31,499 potential outliers. Specifically, there are 1,935 outliers in age, 41 in annual income, and 4,755 in account balance. Additionally, 24,768 outliers are flagged under the churn indicator, which represents the minority class of churned customers.

STATISTICAL ANALYSIS

Statistical evaluation indicates that there are no strong linear correlations among the numerical variables, including annual income, credit score, spending score, and account balance. This lack of correlation suggests that customer financial behaviors and demographic traits act independently. The individual records reveal wide variations, featuring cases of young customers with high spending scores alongside older customers holding substantial balances.

BUSINESS INSIGHTS

The dataset spans key departments such as Sales, Finance, and HR, indicating cross-functional utility. The presence of 24,768 churned records indicates a churn rate of approximately 25% within the transaction history. This high volume of churn highlights an urgent need to segment customers

by department and education level to uncover specific pain points. Understanding why certain segments are departing will allow for more personalized customer engagement.

RISK ASSESSMENT

The primary business risk is the substantial volume of customer churn, which threatens long-term revenue stability. From an analytical perspective, the 6,731 combined outliers in age, annual income, and account balance risk skewing predictive models and business forecasts. Relying on unadjusted data could lead to distorted customer lifetime value estimations and ineffective marketing spend.

AI RECOMMENDATIONS

First, establish data preprocessing rules to treat the numerical outliers in age and account balance before feeding them into predictive models. Second, deploy the recommended scatter and bubble charts to segment customer spending scores against account balances to identify high-value, high-risk cohorts. Third, design targeted retention campaigns specifically tailored to the departments and educational segments experiencing the highest churn rates.

Rule-Based Recommendations

- ✓ ■ No missing values detected.
- ✓ ■ No duplicate rows found.
- ✓ ■■ 31499 potential outliers detected.
- ✓ ■ Histogram, Box Plot and KDE Plot are recommended for numeric columns.
- ✓ ■ Pie Chart, Donut Chart and Count Plot are recommended for categorical columns.
- ✓ ■ Scatter Plot and Bubble Chart are recommended to analyze relationships.

CONCLUSION

The dataset serves as a highly reliable and clean foundation for strategic planning, backed by a 90% health score and zero missing values. By resolving the identified numerical outliers and leveraging targeted data visualizations, the business can transform these insights into proactive

retention strategies, thereby reducing customer churn and improving overall financial performance.